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SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named “SDC2026_TeamName_Preliminary.pdf” before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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1. Team & Project Overview

Team Name	AiX Global
Selected Domain	Domain 2
Selected SDG Track (1–17)	SDG 8 — Decent Work and Economic Growth
Project / Solution Name	Clarity
Name (Leader)	Bhone Myat Paing Soe
Email & Phone (Leader)	jeremywin37@gmail.com ; 0173481780
University Affiliation(s)	Monash University, Sunway University, Univeristy of Nottingham Malaysia
Number of Team Members	4

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

Small and medium enterprises (SMEs) face a constant flood of customer reviews across multiple platforms. For busy owners and managers (in particular those running Malaysian micro-businesses) reading every review, tracking recurring problems, and drafting individual responses is an overwhelming administrative burden. Furthermore, standard rating systems often obscure actionable insights, as a single review might mix specific praise with a critical operational complaint. Valuable feedback is buried, and owners lack the time to trace issues back to their exact source evidence.

This inefficiency directly hampers sustainable economic growth and decent work (SDG 8). When SME owners are bogged down by manual data triage, business quality stagnates. Failing to notice critical operational hazards risks customer safety and damages economic viability.

Our team is tackling this challenge by shifting the SME workflow from administrative busywork to actionable business improvement. Our project transforms unstructured, mixed feedback into clear, evidence-backed operational tasks. By empowering local business owners to quickly move from a customer complaint to an approved, verifiable repair, the application relieves the digital paperwork burden. This enables local enterprises to maintain safe environments, operate efficiently, and focus on sustainable growth.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

Clarity is a web application that transforms unstructured customer feedback into evidence-backed operational workflows for small and medium enterprises, relieving owners of administrative busywork to promote productive, decent work environments. Instead of forcing owners to read every generic comment, a multi-stage pipeline first screens incoming reviews to identify only those containing specific, actionable details. A Jev classification model performs this initial screening; however, if Jev fails or is uncertain, an OpenAI model acts as a fallback to screen the review using the exact same rules. When a customer leaves a mixed review, the software uses OpenAI to separate these points into distinct observations while strictly preserving the customer's exact words for evidence. As new feedback arrives, the system automatically connects related observations, alerting the manager to recurring, branch-specific issues.

Once an issue is identified, an owner can propose a task and assign it to an employee, who can start work and submit a report. Crucially, the owner must then enter a separate verification note before the system considers the task fully resolved. Finally, the application uses this verified note to prepare individualized, truthful responses for the affected customers. If an owner resolves one issue but the source review contains other unresolved complaints, the AI intelligently attaches a partial-reply warning to the draft. This warning restricts batch approval and requires explicit, individual owner sign-off to ensure accurate customer communication.

Leveraging core skills gained from technical workshops, the application relies on GenAI APIs such as a typed Jev adapter to maintain reliable, server-side model interactions. The system is built on a robust database architecture using PostgreSQL and Drizzle (ORM). Finally, the full-stack Next.js application is constructed with clear environment configurations for seamless cloud hosting and deployment on platforms like Replit.

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

The prototype is built as a unified, full-stack Next.js web application utilizing TypeScript and the App Router for seamless server and client integration. The user interface relies on standard React components and a highly modular, token-based CSS architecture. Visual styles are managed through specific files like tokens.css and workspace.css, ensuring that visual design is strictly separated from the React domain logic. For robust data management, the application employs a PostgreSQL database accessed through a modern ORM, utilizing runtime validation with Zod at all external input boundaries to maintain data integrity.

The system architecture follows a server-side data flow designed for strict data isolation and reliability. The entire application is strictly scoped by a business identifier across all API routes and data tables, allowing the same database to handle distinct entities simultaneously without data leakage. Database migrations, such as the hotel workflow extension, successfully expand the schema to support complex employee roles, verification notes, and recurring issue tracking without disrupting existing records.

External customer feedback enters the application layer, which persists the raw data in the database before routing it through the AI pipeline. The server orchestrates requests to the TypeSafe Jev API for initial screening and the OpenAI API for fallback screening, structured observation extraction, and automated reply drafting. All AI calls are executed strictly server-side to secure API keys. Finally, the project leverages Replit's cloud hosting platform, utilizing standard configuration variables and Replit’s managed database environments for smooth and reliable deployment.

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

Our team applied design-thinking by centering the architecture around the practical constraints of a busy SME manager. To ensure user-friendliness and technical viability, we strictly separated the visual presentation from the core business logic. By utilizing a modular, token-based CSS architecture, we created accessible, responsive interfaces that can be rapidly iterated without risking the underlying workflows. Furthermore, the system builds user trust by preserving the exact source wording of all customer feedback and requiring explicit human verification before resolving tasks or drafting replies.

For deployment readiness, we engineered a scalable backend where the PostgreSQL database is strictly scoped by business ID. This multi-tenant design allows a single deployment to securely support multiple businesses, such as hotels or salons, simultaneously without data crossover. The codebase is fully prepared for Phase 2 cloud deployment, utilizing standard environment variables, repeatable database migrations like the hotel workflow extension, and a simulated Orbyte handoff that proves the application can cleanly integrate into a real-world business ecosystem.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	**Your Response**
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	Clarity primarily addresses UN SDG Target 8.3: "Promote development-oriented policies that support productive activities, decent job creation, entrepreneurship, creativity and innovation, and encourage the formalization and growth of micro-, small- and medium-sized enterprises" (United Nations Department of Economic and Social Affairs [UN DESA], n.d.). Alongside this, the application advances Target 8.2: "Achieve higher levels of economic productivity through diversification, technological upgrading and innovation" (International Labour Organization [ILO], n.d.). By equipping labor-intensive SMEs—such as hotels, restaurants, and salons—with a strictly business-scoped AI architecture, Clarity automates initial feedback screening and exact-substring evidence extraction. By tracking verified employee task resolutions and drafting accurate, partial-reply-aware customer responses, Clarity replaces manual administrative busywork with scalable operational productivity.

Current Problem Baseline *What is the starting condition of the problem, expressed a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	In the hospitality and SME sectors, managers face a significant administrative burden when managing online reputation, leading to low response rates and lost operational insights. Academic research indicates that a substantial portion of customer feedback remains unaddressed; for instance, an empirical study found that, on average, only 31.5% of TripAdvisor reviews received a management response (Proserpio & Zervas, 2017). More recent analysis demonstrates that even 5-star hotels leave nearly 40% of reviews unaddressed, struggling with the time required to manage both positive and negative feedback (Sufi et al., 2025). This low response rate is exacerbated by the fact that overall star ratings obscure mixed feedback, burying recurring operational or safety hazards inside unstructured text. Because poor time management and inefficiency are identified as primary barriers that hinder the progression and expansion of Malaysian SMEs (Exploring the Challenges and the Implementation of Lean Practices, 2023), managers are currently losing valuable hours parsing generic noise rather than addressing specific operational hazards.
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	Advancing UN SDG Targets 8.2 and 8.3, our goal for future real-world validation is to track the percentage reduction in manager time spent triaging feedback and preparing replies, alongside the percentage of resolved operational issues with verified, ready-to-post customer responses. While the current hackathon prototype relies exclusively on synthetic data and does not natively calculate or display these metrics yet, our business-scoped architecture lays the necessary groundwork. The system's append-only activity log successfully records the complete cycle from initial AI screening and exact-substring observation extraction to employee task reporting and final owner verification notes. Once deployed with live businesses, this key performance indicator will utilize these logs to measure the definitive shift from manual administrative reading to verifiable operational action, tracking exactly how many issues are successfully processed through to a localized reply draft.

Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	By the Phase 2 prototype presentation, we target a 40% reduction in review triage and response preparation time. This efficiency gain is designed to help businesses comfortably meet the academic and industry best practice of responding to customer reviews within 24 hours (Madnick et al., 2017). Concurrently, we target 100% evidence traceability for grouped operational themes and an 80% or higher response-readiness rate for verified resolved tasks. This is achieved by filtering out generic noise during screening, automatically flagging unresolved multi-issue complaints with partial-reply warnings, and enabling the batch approval of clean reply drafts.

References

Exploring the Challenges and the Implementation of Lean Practices. (2023). Jurnal Pengurusan. https://www.ukm.my/jurnalpengurusan/wp-content/uploads/2023/03/jp_67-8.pdf

International Labour Organization. (n.d.). Sustainable Development Goal 8: Decent work and economic growth. https://www.ilo.org/topics-and-sectors/decent-work-and-2030-agenda-sustainable-development/sustainable-development-goal-8-decent-work-and-economic-growth

Madnick, S., et al. (2017). *Management Responding Strategy to Customer Online Reviews*. Massachusetts Institute of Technology. https://web.mit.edu/smadnick/www/wp/2017-11.pdf

Proserpio, D., & Zervas, G. (2017). *Online Reputation Management: Estimating the Impact of Management Responses on Consumer Reviews*. Boston University. https://people.bu.edu/zg/publications/review-responses.pdf

Sufi, T., Goyal, A. K., & Chaudhary, H. K. (2025). How hotels respond to online ratings. In L. C. Carvalho, L. Reis, A. M. N. A. B. Figueiredo, & N. Russo (Eds.), *Leveraging technology for organizational adaptability* (pp. 285–303). IGI Global. https://doi.org/10.4018/979-8-3693-6045-3.ch012

United Nations Department of Economic and Social Affairs. (n.d.). Goal 8 | Department of Economic and Social Affairs. https://sdgs.un.org/goals/goal8

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

- [x] All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.
- [x] The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
- [x] The proposed solution targets a functional, deployed application rather than a theoretical business case study.
- [x] All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☑ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☑Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
Next.js, React, React DOM	Frameworks and libraries	MIT
Drizzle ORM	Database library	Apache 2.0
OpenAI SDK	API client library	Apache 2.0
PostgreSQL pg driver, Zod, server-only	Libraries	MIT
TypeScript, Vitest, ESLint, Prettier	Development and test tools	Apache 2.0 or MIT, depending on package
OpenAI GPT-5 mini	Hosted AI model, used in live mode	OpenAI service terms
Jev by Typesafe	Hosted review-screening AI(model), used in live mode	Typesafe service terms
Clarity review fixtures	Synthetic dataset created by our team	Team-created; no third-party dataset

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☑ No

If yes, please describe the nature of the agreement and which member it concerns:

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.

[Signature]



[Date] 10/25/2026